Norwegian Meteorological Institute

RAD320: Marine Radioecology

Part I

- **Physical environment**
- **Modeling**

Magne Simonsen, Norwegian Meteorological Institute

29.10.2026

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Plan 2026 Radionuclides in the marine environment

Day 1 Thursday 29. Oct

1. Lectures
    a. Physical marine environment
    b. Modeling transport of radionuclides in the marine environment
2. Hands-on introduction to OpenDrift
    a. Test that installation at desktops works satisfactorily
    b. Simulate surface drift

Day 2 Thursday 5. November

1. Lectures
    a. Radionuclides in the marine environment
    b. Biological uptake
    c. Future Challenges
2. OpenDrift Exercises
    a. Simulate radionuclide drift
    b. Compute concentrations

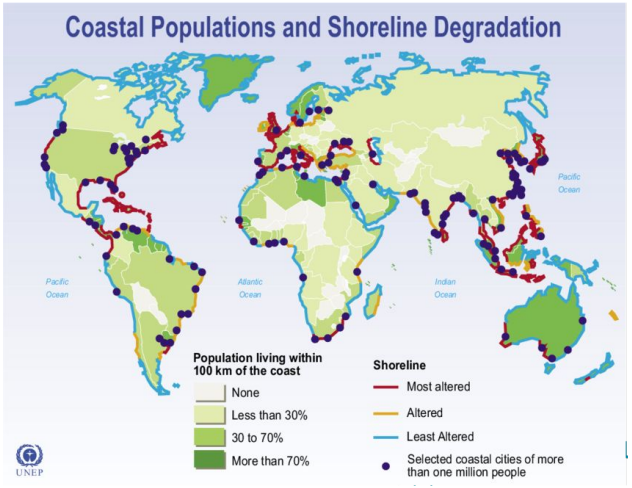
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Physical marine environment

The global marine environment

- The oceans cover 71% of Earth's surface
- Contain 97% of the global water
- 50-80% of all life is found in the oceans
- 2.5 billion people live in the near-coastal zone
- Oceans are home to a huge variety of marine life, including more than 230,000 known species, probably many more to be discovered

- Great variety of ecosystems, from Arctic to the tropics
- Key role in regulating climate
  - sink for carbon
  - heat uptake and distribution

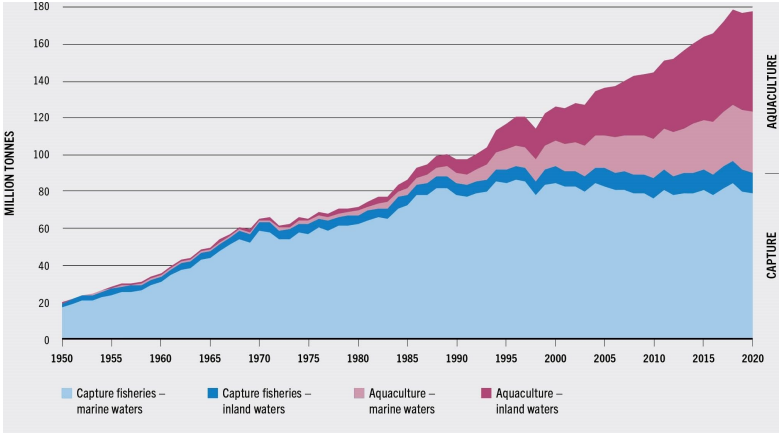


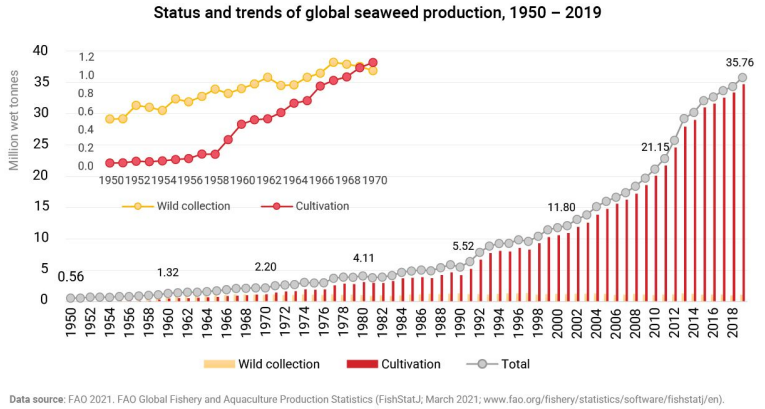
Source: Burke et al., World Resources Institute, Washington DC, 2001; Paul Harrison and Fred Pearce, *AAAS Atlas of Population and Environment 2001*, American Association for the Advancement of Science, University of California Press, Berkeley.

The oceans as a resource

- Fisheries
- Algae, seaweed
- Minerals
- Energy
- Oil and gas
- Tourism and recreation
- Shipping and transport
- Waste dumping







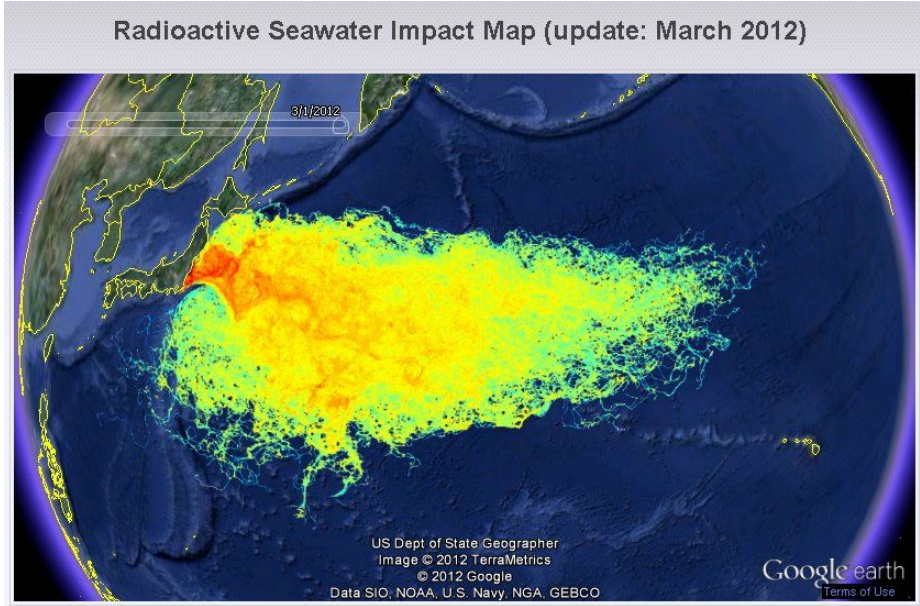
Data source: FAO 2021. FAO Global Fishery and Aquaculture Production Statistics (FishStatJ; March 2021; www.fao.org/fishery/statistics/software/fishstatj/en).

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Ocean currents

- Main drivers for radionuclide transport in the marine environment

- Large scale ocean currents
- Coastal currents at regional scales
- Local features close to the shore



https://sites.suffolk.edu/lmeuse/2014/10/24/fukushima-daiichi-nuclear-disaster/

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Ocean circulation

- **Ocean currents are**
- Flow of water masses
- “Wind in the ocean”

- **Driving forces**
- Pressure gradients
  - Sea level height
  - Density (function of temperature and salinity)
- Wind stress

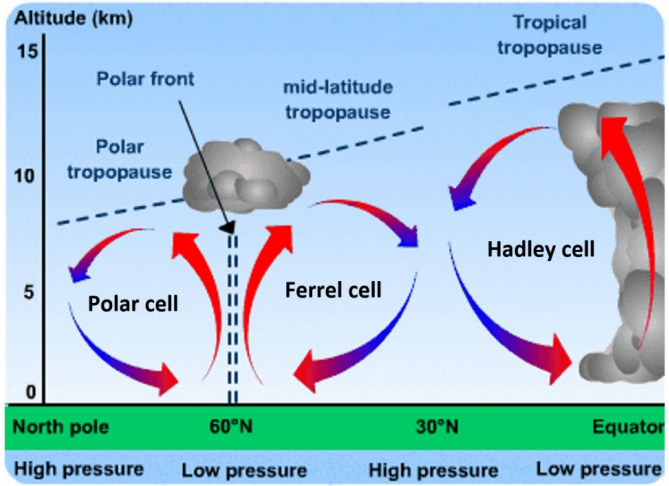
- **Other factors**
- Coriolis acceleration
- Bottom topography
- “Western boundary currents”: Gulf Stream



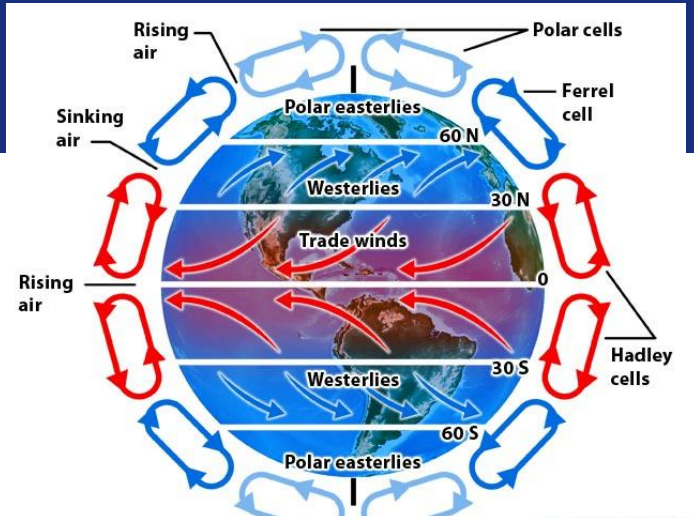
https://courses.seas.harvard.edu/climate/eli/Courses/misc/201209-Mallorca/Sources/2-THC/from-EPS131/2-RAPID/gulf_stream.php.html

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Atmospheric Circulation

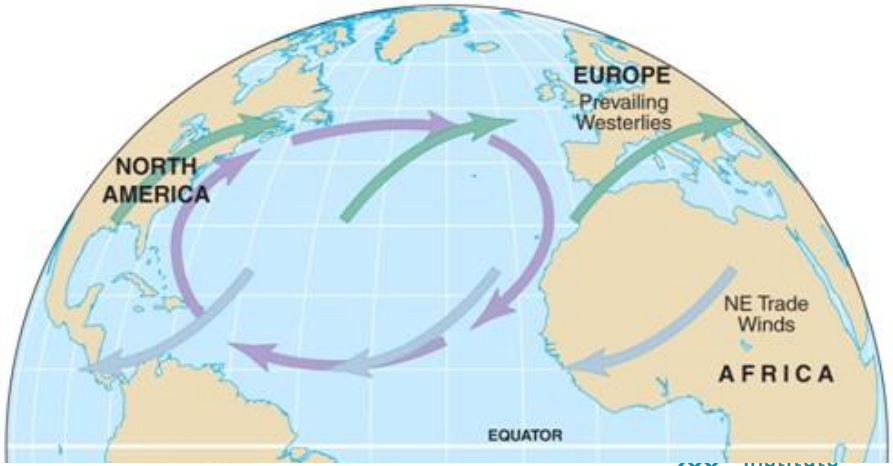


https://content.meteoblue.com/ro/meteoscool/large-scale-weather/atmospheric-circulation



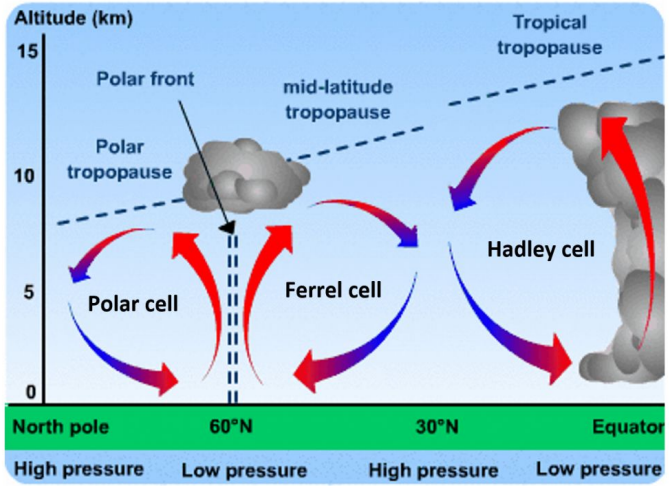
https://picswe.net/pics/convection-currents-wind-78.html

- Uneven heat flux from the sun cause the large scale global wind circulation patterns
- Wind generate drag at the ocean surface
- Clockwise circulation in the North Atlantic basin

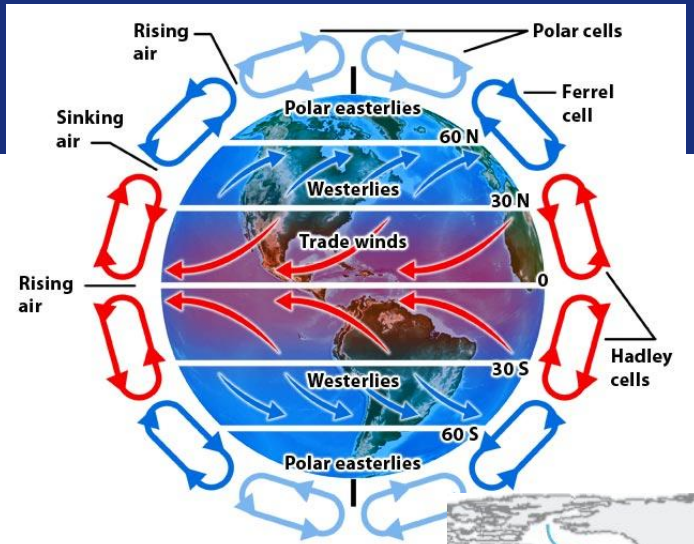


Pearson Education Inc, 2011

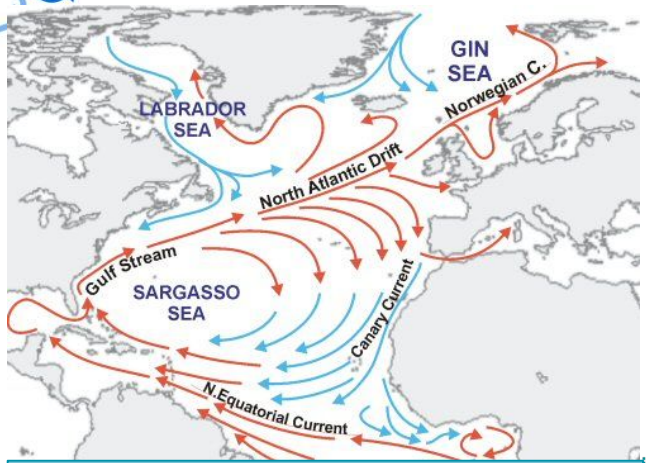
Atmospheric Circulation



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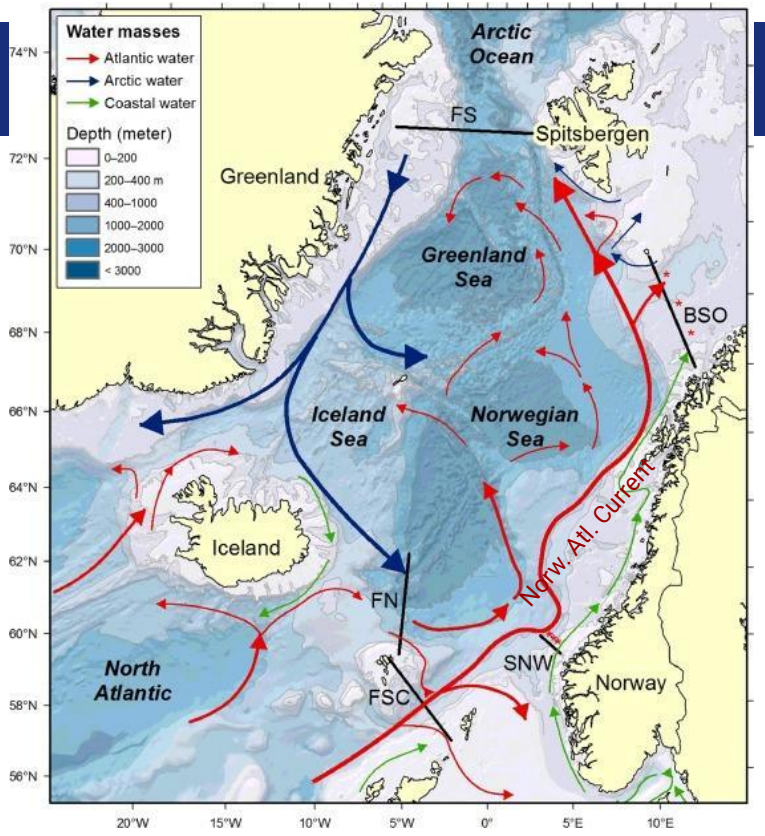
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Norwegian Atlantic Current

- Extension of the Gulf stream
- Important for the Northern Europe climate

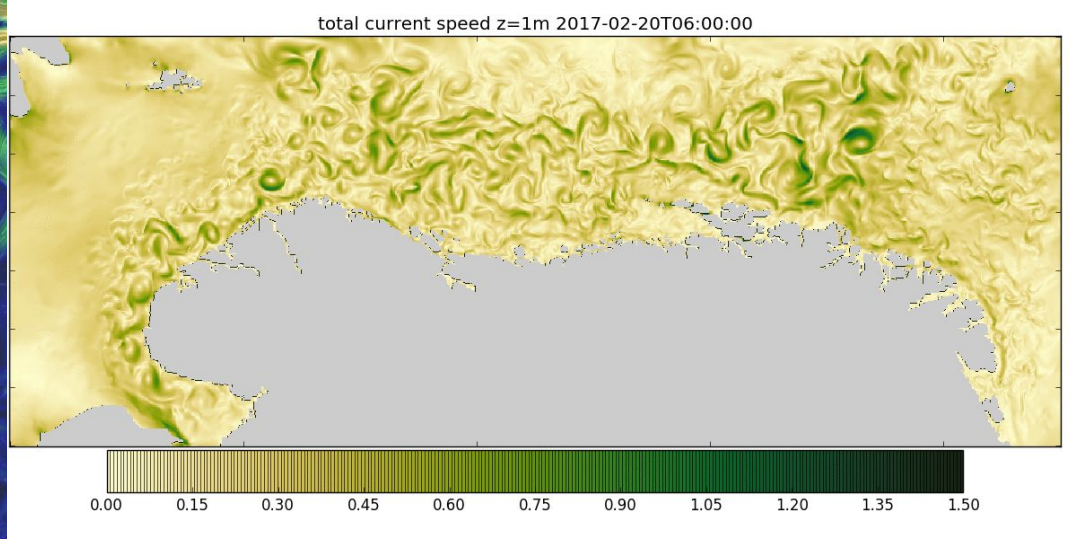
Main characteristics

- Weak stratification
- Topographically steered, follows the continental slope
- Eddy generation
  - Across-slope transport to the Lofoten Basin
- Branches to Arctic Ocean
  - Barents Sea Opening
  - Fram Strait (West Spitsbergen Current)
- Continuous heat loss



Ocean currents are NOT stationary “river-like” features, but have large variability in time and space

(earth.nullschool.net)



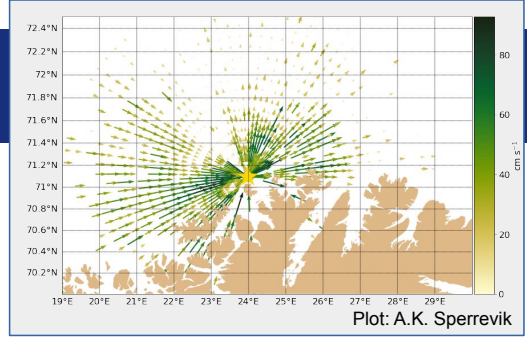
earth

Observing the currents

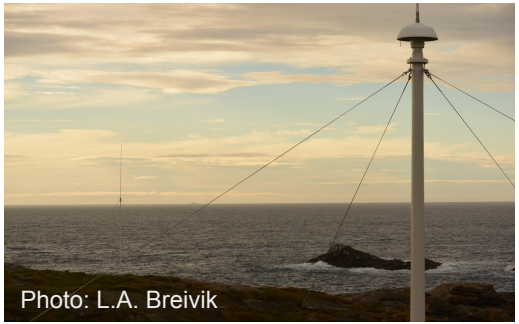
Many different observation instruments are available

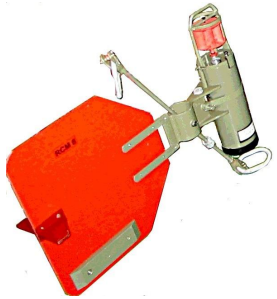
- Moored current meters
- Drifters / floats
- Satellites: currents derived from surface elevation
- HF radar
- ADCP

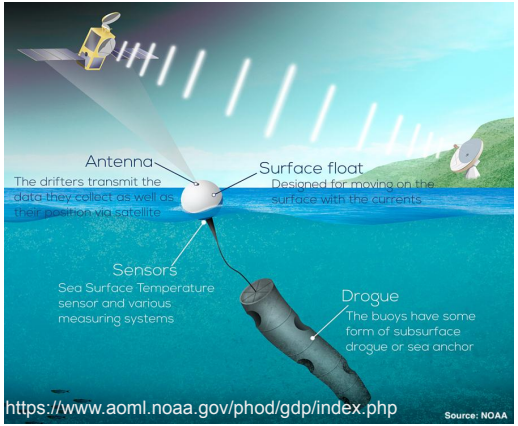
But due to the high variability and the scarce observation network, a full monitoring and prediction of the ocean state is difficult today

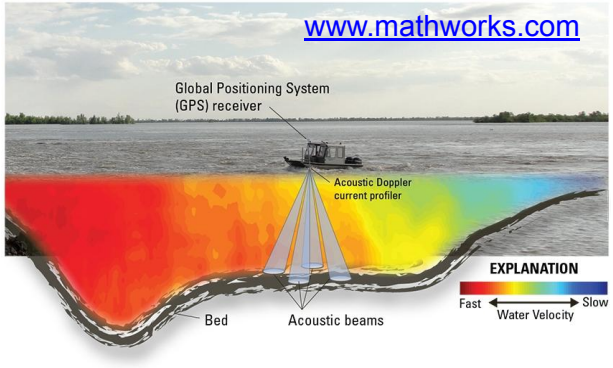


Plot: A.K. Sperrevik


Photo: L.A. Breivik







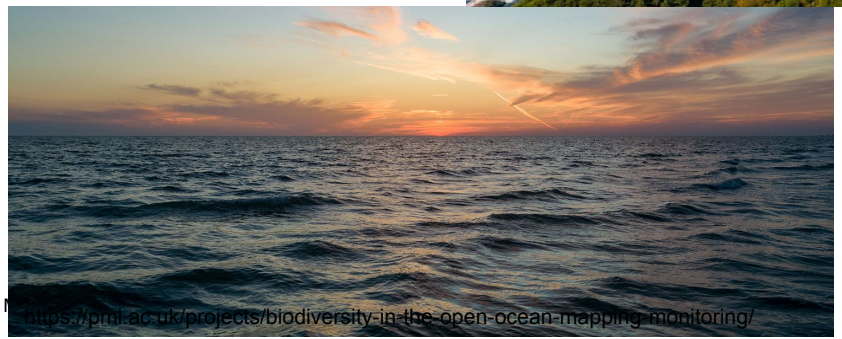
The different marine environments

- Estuaries
- Coasts
- Open Ocean



(wikipedia)



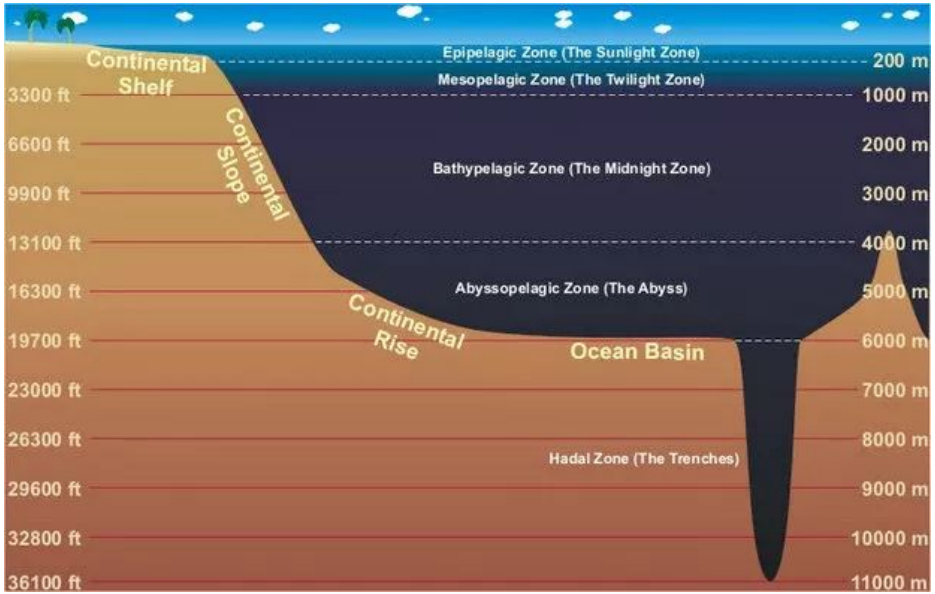


https://pml.ac.uk/projects/biodiversity-in-the-open-ocean-mapping-monitoring/

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Open Ocean environment

- Wind-driven circulation
- Large depths
- Lower productivity than in coastal zones
- Non-surface interior
  - Low light penetration
  - High pressure
  - Low temperature
  - Specialized organisms
- Dominated by
  - Pelagic species
  - Plankton
  - Large fish and mammals



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Coastal environment

- Shelf break separates the coasts from the open oceans
- Land-ocean-biota interactions
  - Main habitat for marine species - high biodiversity and biological productivity
  - Ecosystems
- Characteristics:
  - Shallow water
  - Waves, tides, rivers, topography
- Mixing of different water masses
  - Strong gradients
  - Strong stratification
  - Large variability
- High particle concentration
- High population density - Human impact on the environment
- Contaminants

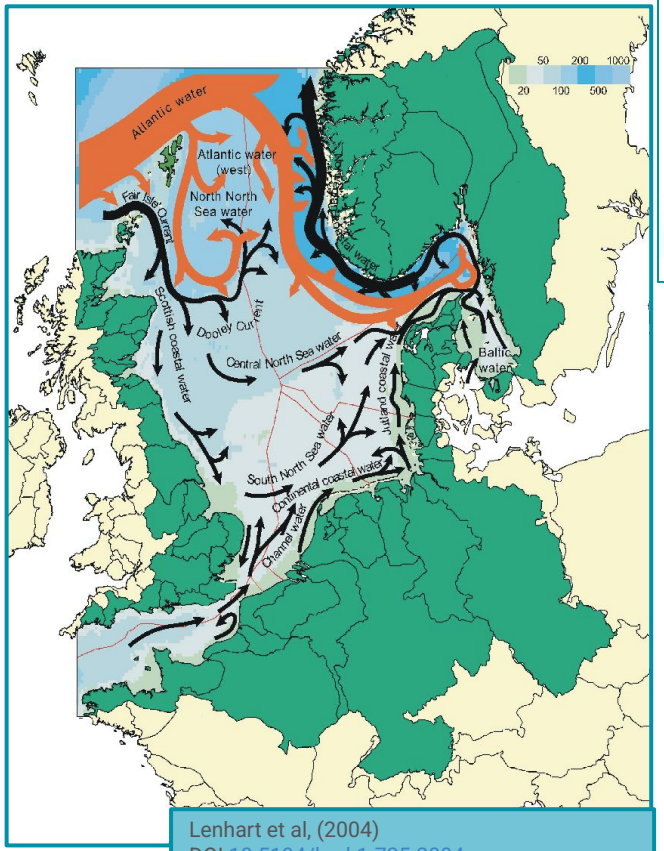




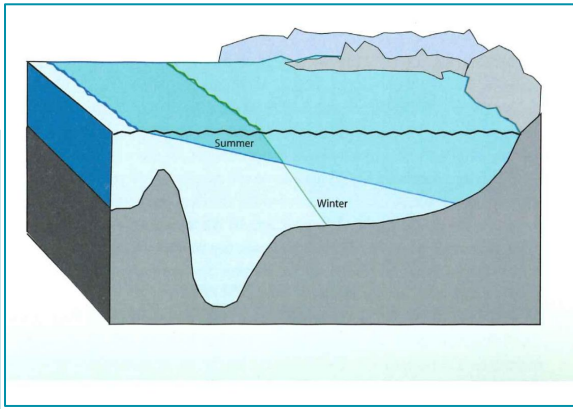
Coastal currents

Coastal currents

- Freshwater from rivers
- Summer: warmer than the offshore water
- winter: colder than the offshore water
- Follows the coast with land on the right hand side



Lenhart et al, (2004)
DOI:10.5194/bgd-1-725-2004

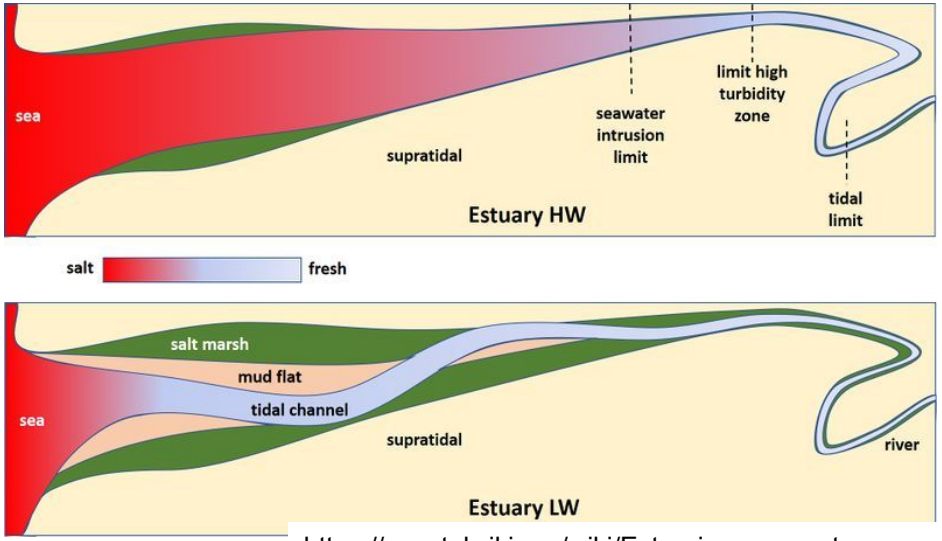


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Estuarine environment

- Where the river water meets the coastal water
- Among the most productive environments in the world
- Important characteristic:
  - physically confined
  - mix of seawater and freshwater
  - strongly affected by tides
  - habitat for species that are naturally tolerant to stresses caused by temporal and spatial variations in salinity, temperature, dissolved oxygen, turbidity, and bottom sediment

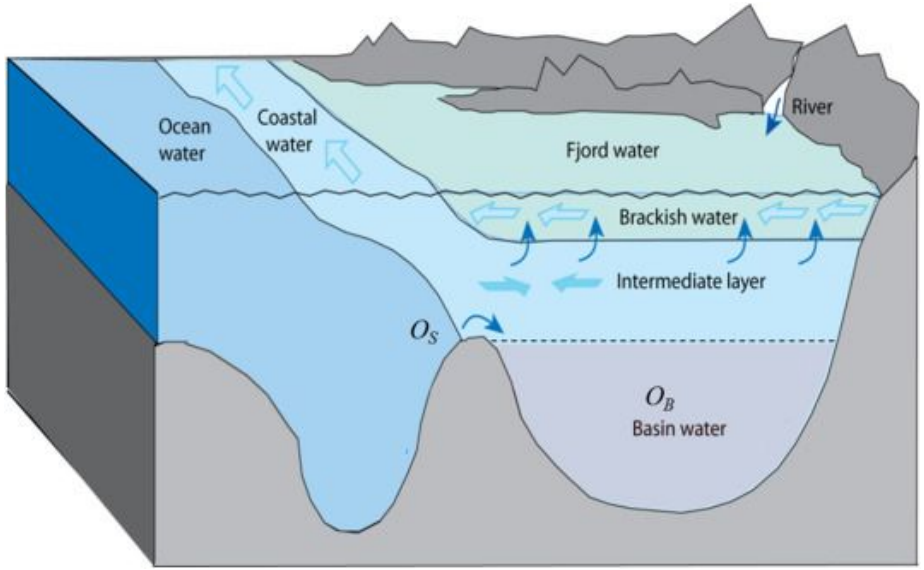




https://coastalwiki.org/wiki/Estuarine_ecosystems

Fjord circulation

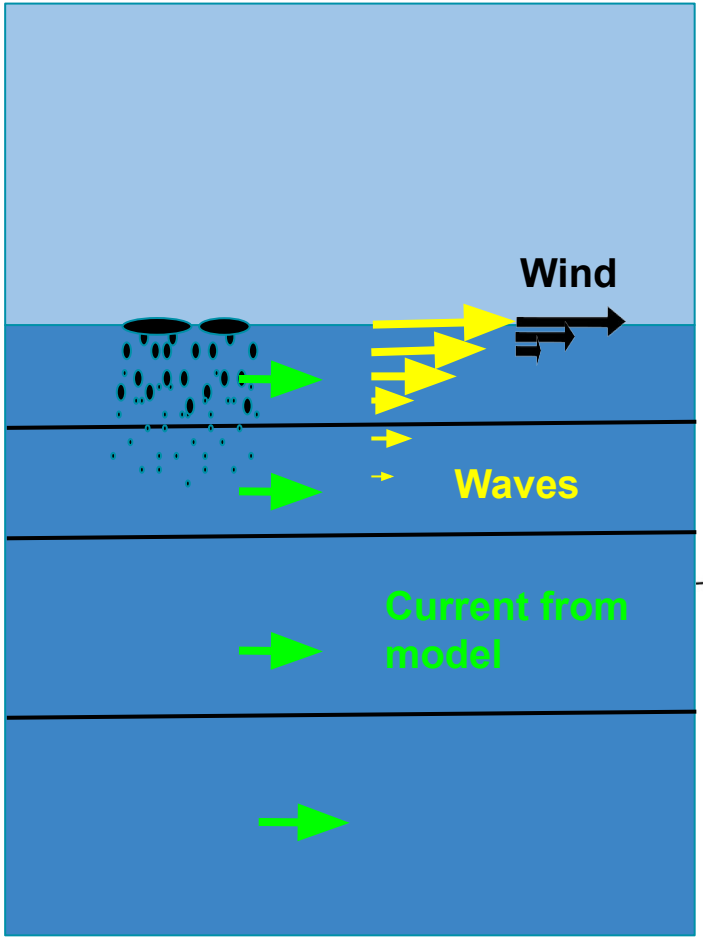
- Upper layer
  - Low salinity (low density)
  - General flow out of the fjord
  - Currents are affected by wind
- Intermediate layer
  - Extends down to the sill depth
  - Higher salinity
  - Mixes with the upper layer
  - Horizontal exchange with coastal water
- Basin water
  - Low exchange with surrounding water
  - Weak currents
  - High density
  - Low oxygen

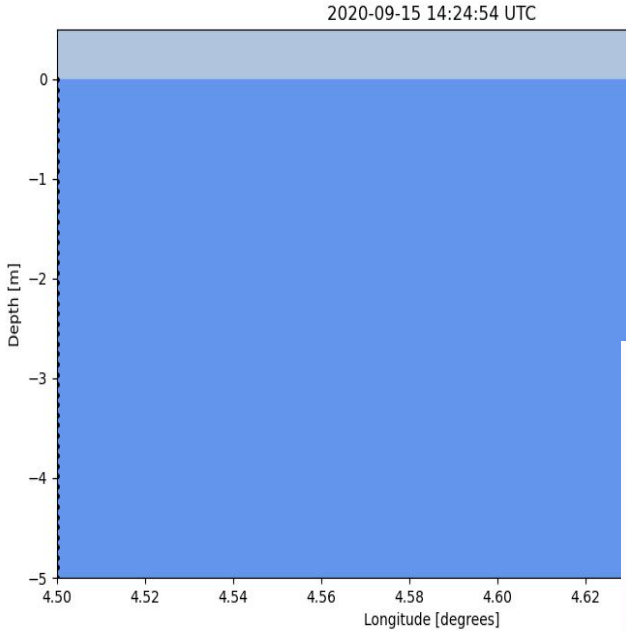


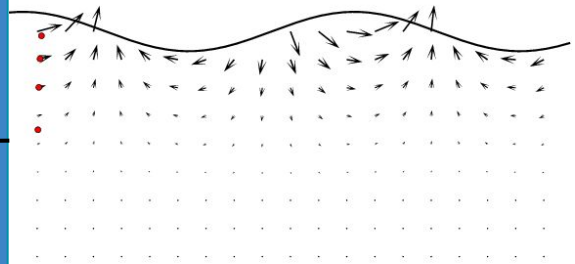
Aksnes et al (2019)
https://doi.org/10.1016/j.ecss.2019.106392

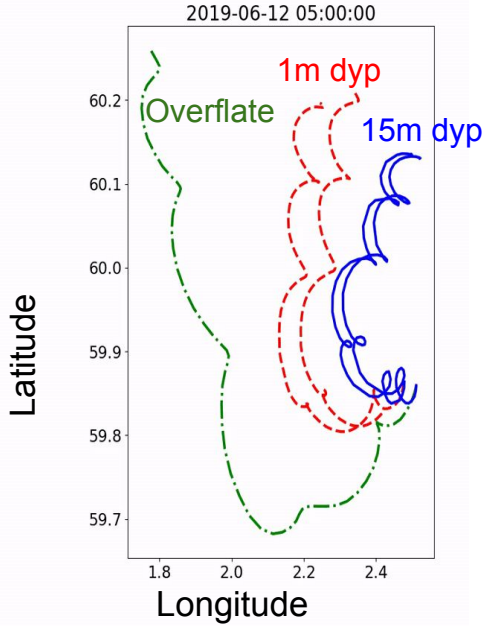
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Vertical variation





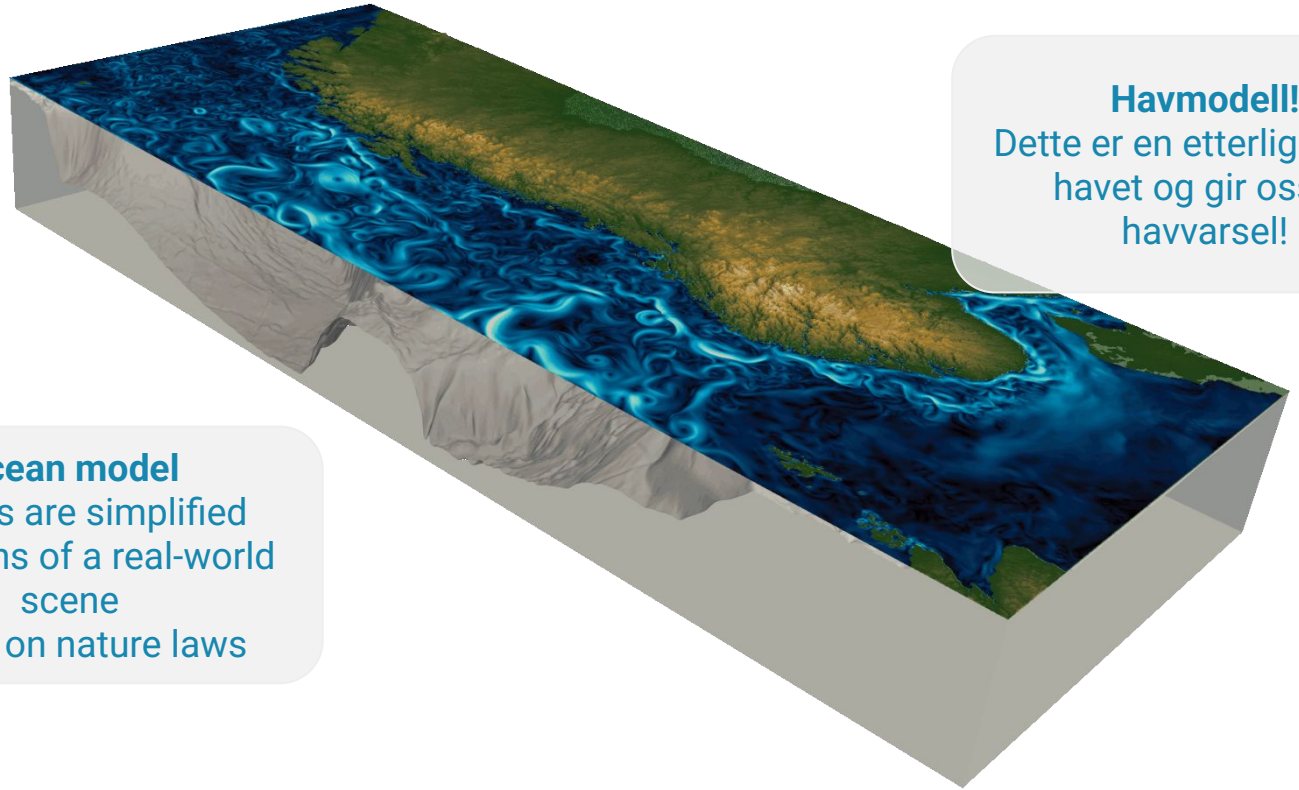




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Modeling transport of radionuclides in the marine environment

What is an ocean model?



Havmodell!
Dette er en etterligning av havet og gir oss et havvarsel!

Ocean model
Models are simplified imitations of a real-world scene
Based on nature laws

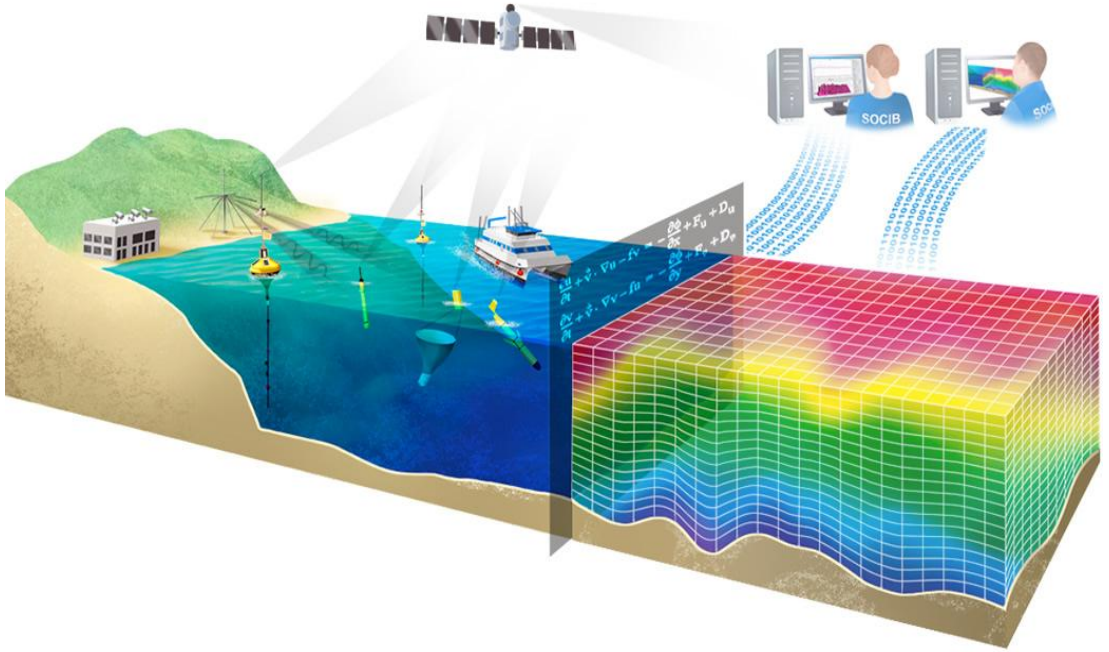
Numerical radionuclide transport models

Models are commonly used to:

- estimate contaminant transport where observations are limited
- predict future and hypothetical scenarios
- open up for process investigation

Model results are affected by

- Assumptions
- Simplifications
- Parameterizations



medclic.es

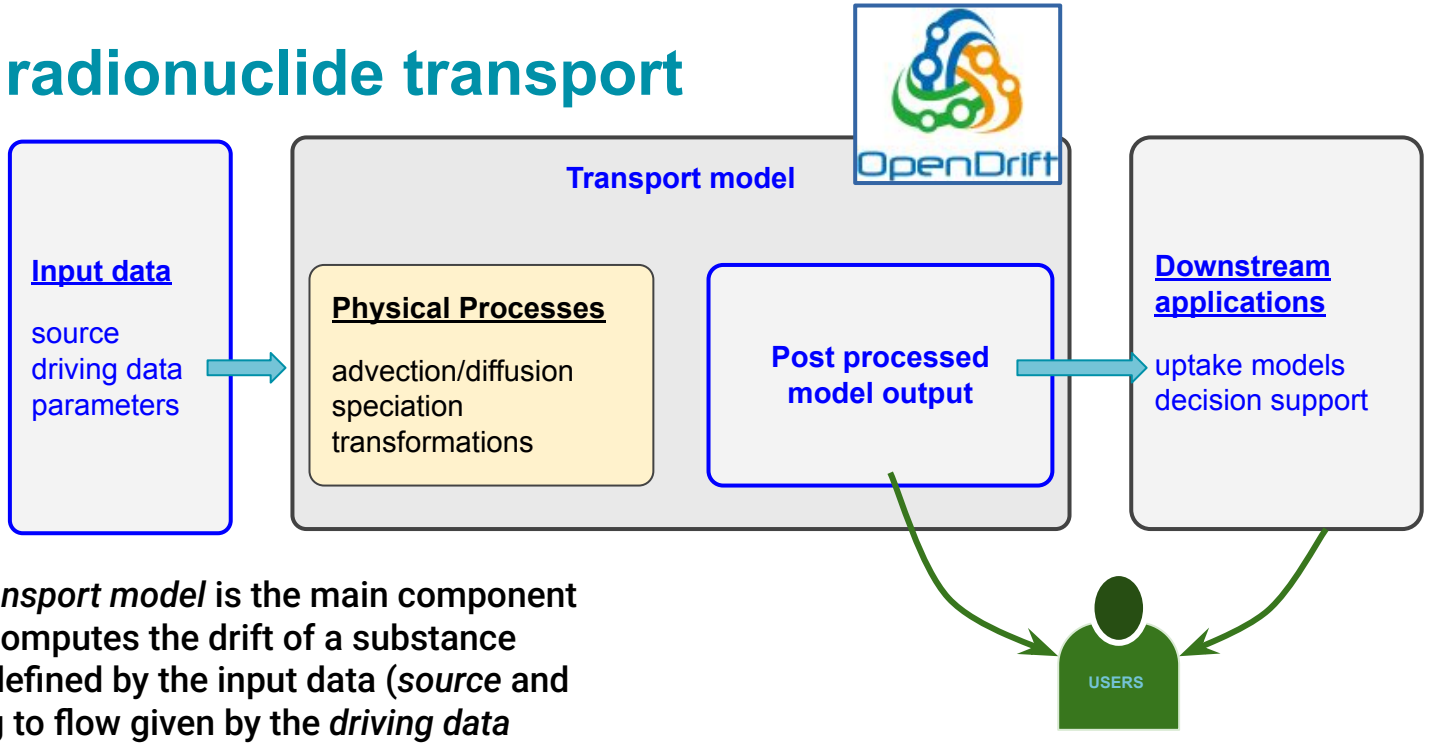
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Trade-off in model design

Models are trade-off between being

- Realistic
  - Include process-based algorithms based on nature laws (need of parameters)
- General
  - Applicable to a variety of similar situations and systems (useful for forecasting)
- Precise
  - Accurate reproduction of measurements
  - Site- and case-specific
  - Empirical tuning
- Fast
  - Computationally efficient

Modelling the radionuclide transport



- In this context, the *Transport model* is the main component
- The transport model computes the drift of a substance (here: radionuclides) defined by the input data (*source* and *parameters*) according to flow given by the *driving data*
- The drift can follow the flow passively (*advection and diffusion*), or be be modified by some sort of *physical process*
- Model results need usually to be *processed* to be useful for interpretation/analysis or *downstream applications*

Modelling the radionuclide transport






https://ariabeniz.com/title-post/
Getty Images/iStockphoto
https://bellona.org/

Input data

source
driving data
parameters

Transport model

Physical Processes

advection/diffusion
speciation
transformations

Post processed model output

Downstream applications

uptake models
decision support

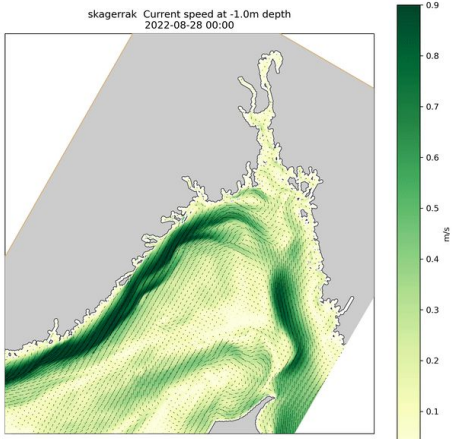
OpenDrift

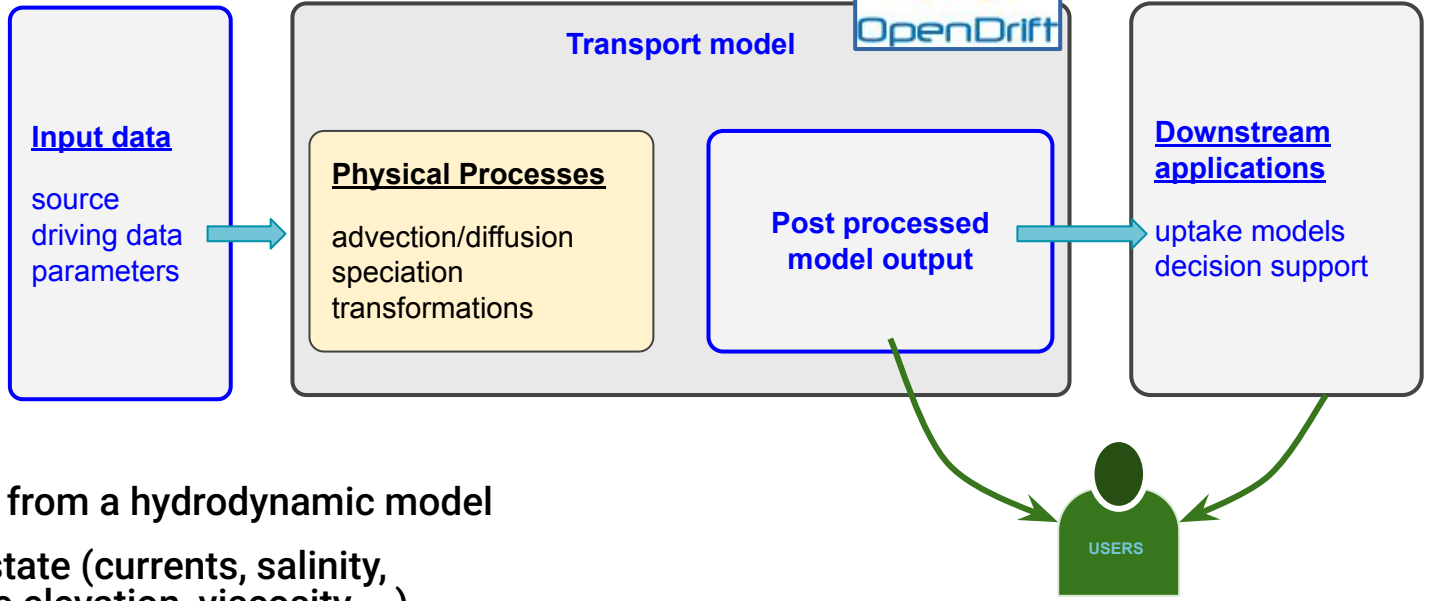
USERS

Examples of **sources** can be

- atmospheric deposition from test detonations
- accidents at nuclear installations
- operational discharges
- releases/accidents from nuclear powered vessels
- disposed nuclear waste
- …

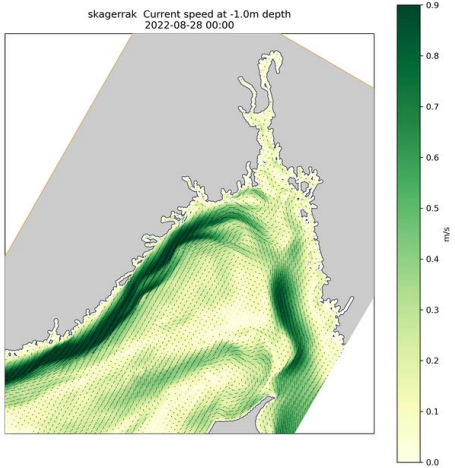
Modelling the radionuclide transport





- Input: driving forces from a hydrodynamic model
- 3D Physical Ocean state (currents, salinity, temperature, surface elevation, viscosity, …)
- operational forecasts (short term) + archives
- Typically hourly output data, but often at lower temporal resolution
- Key: horizontal resolution and domain coverage

Modelling the radionuclide transport

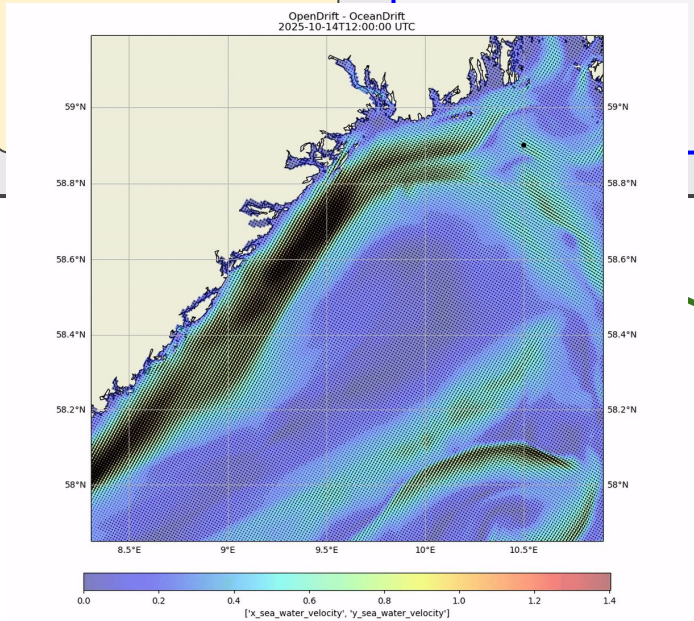


Input data

source
driving data
parameters

Transport model

Physical Processes



Downstream applications

uptake models
decision support

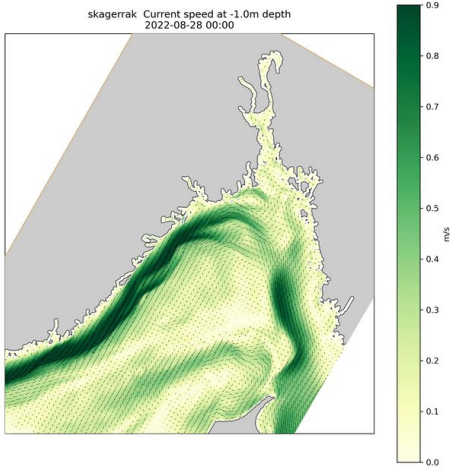
USERS

Transport model

Compute particle motion according to local currents + specific behavior (speciation, transformations)
Common usage:

- Emergency preparedness
- Compute transport from well-known/well-defined sources at timescales days-weeks

Modelling the radionuclide transport



Input data

source
driving data
parameters

Transport model

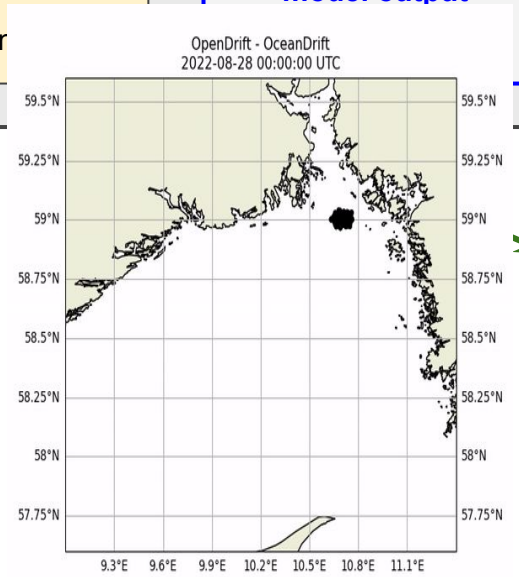
Physical Processes

advection/diffusion
speciation
transformation

Post processed model output

Downstream applications

uptake models
decision support



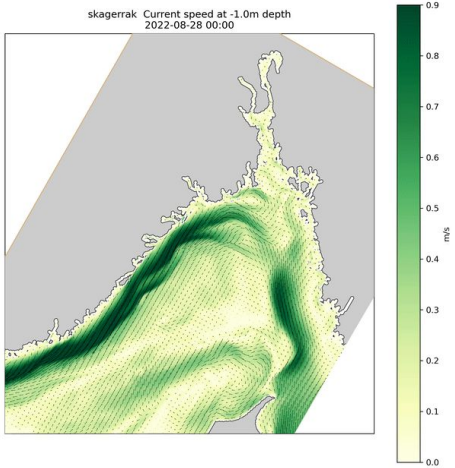
USERS

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Modelling the radionuclide transport



Input data

source
driving data
parameters

Transport model

Physical Processes

advection/diffusion
speciation
transformations

Post processed model output

Downstream applications

uptake models
decision support

USERS



User-friendly output

- Output needs processing to be useful
- Concentrations
- Time series
- Decision support systems
- Environmental impact: Uptake models

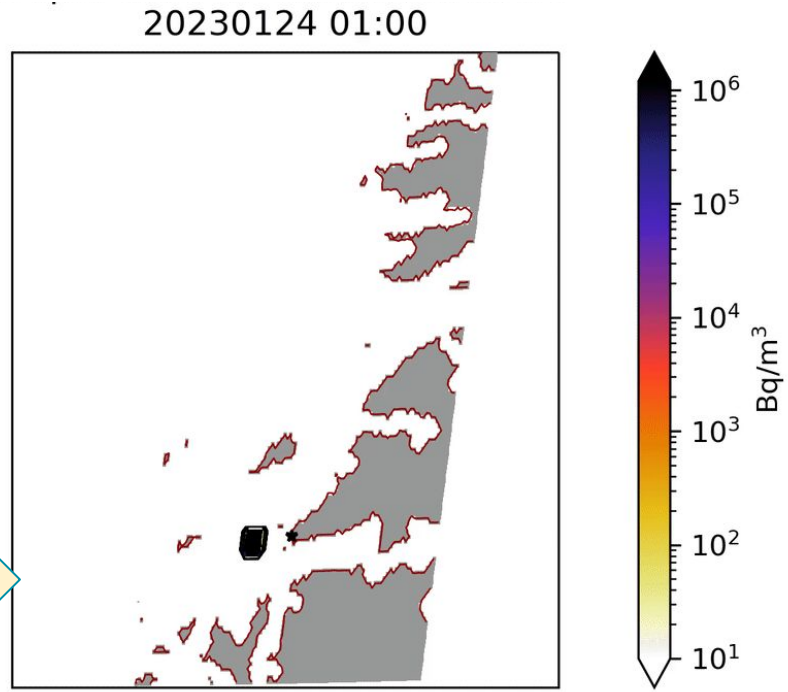
Example of a nuclear preparedness exercise

- Discharge outside Bodø
- 7 days simulation
  - real ocean weather on 24.-31. January 2023

- Output:
  - Hourly maps of activity concentration in surface
  - Time series ⇒ uptake in fish

Total surface concentration

- **LMM species**
- Particle-bound species
- Sediment species

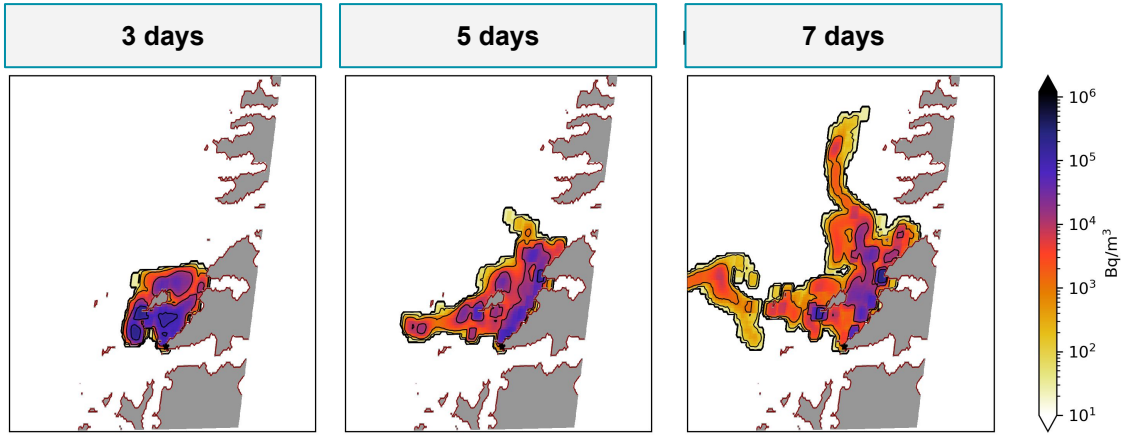


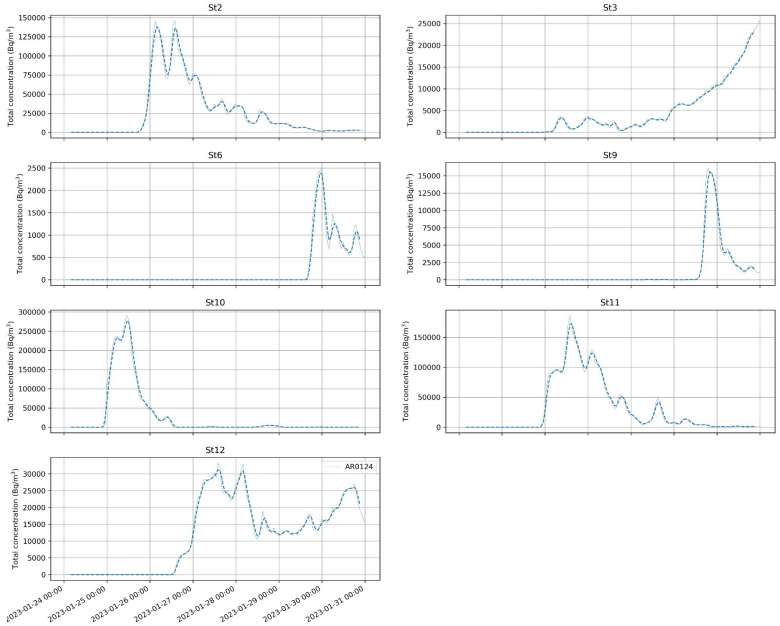
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Example of a nuclear preparedness exercise

- Discharge outside Bodø
- 7 days simulation
  - real ocean weather on 24.-31. January 2023

- Output:
  - Hourly maps of activity concentration in surface
  - Time series ⇒ uptake in fish

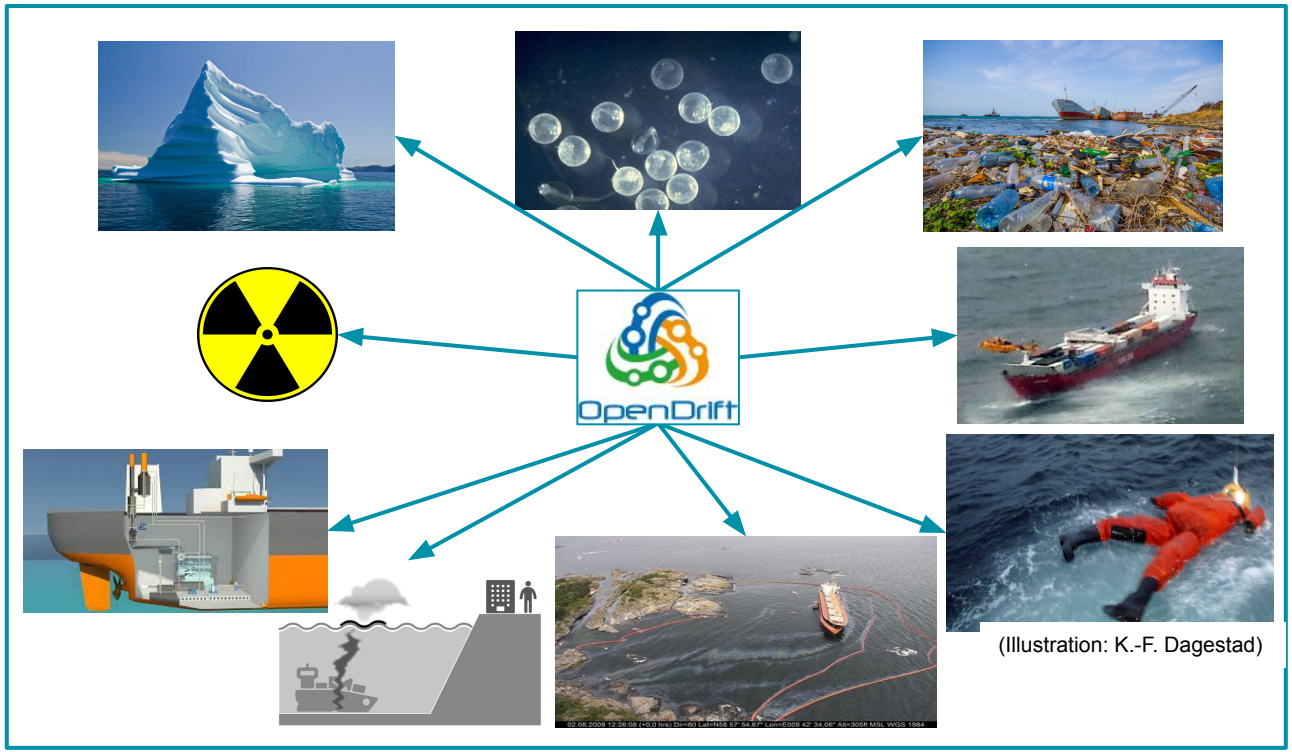




Norwegian Meteorological Institute

OpenDrift model

- Transport model
- Generic multi-modular **Lagrangian trajectory model** framework
- Operational preparedness
- Radionuclide module
- open source: www.opendrift.github.io



(Illustration: K.-F. Dagestad)

Lagrangian model:

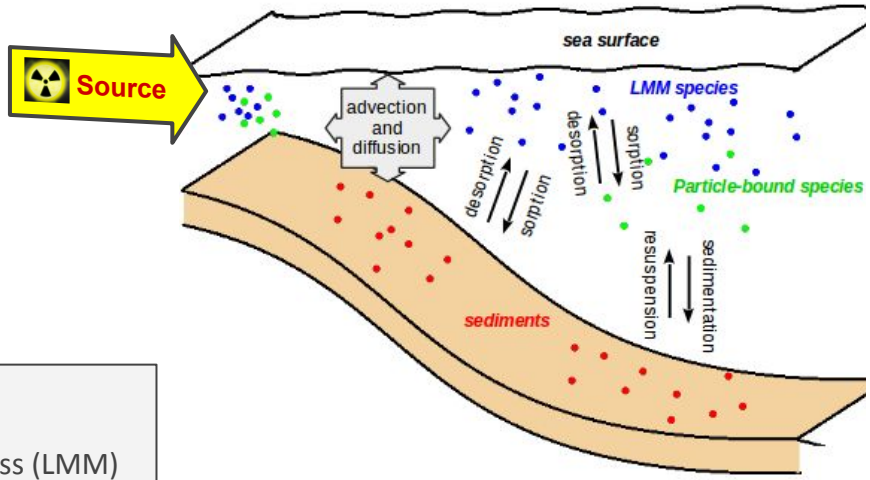
- Evaluate the changes of a water parcel which follows the mean flow
- Compute trajectories of a finite number of independent *numerical particles*

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OpenDrift transport model

The **radionuclide module** takes into account

- Horizontal and vertical advection and diffusion with the currents
- Speciation
- Dynamic transformations between the species
- Settling of particle species
- Sedimentation/resuspension at seabed
- Slowly reversible, irreversible particle species



Species

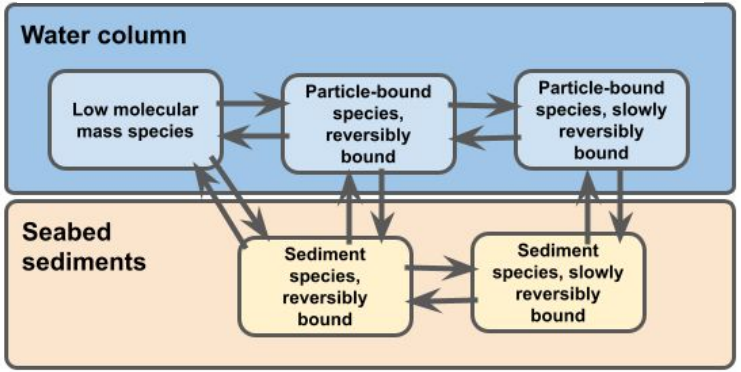
- Low Molecular Mass (LMM)
- Colloidal
- Particle-bound
- Sediment-bound

Speciation affects

- transport
- biological uptake

Transformation processes

- sorption / desorption
- sedimentation/resuspension

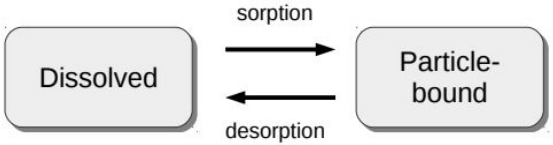


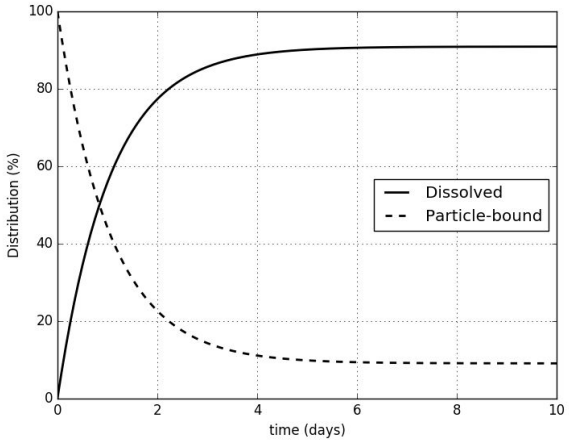
Radionuclide transport model: Dynamic transformations

- In a closed system with two species (A_w, A_s), the time evolution of distribution of radionuclide activity will follow this equation:

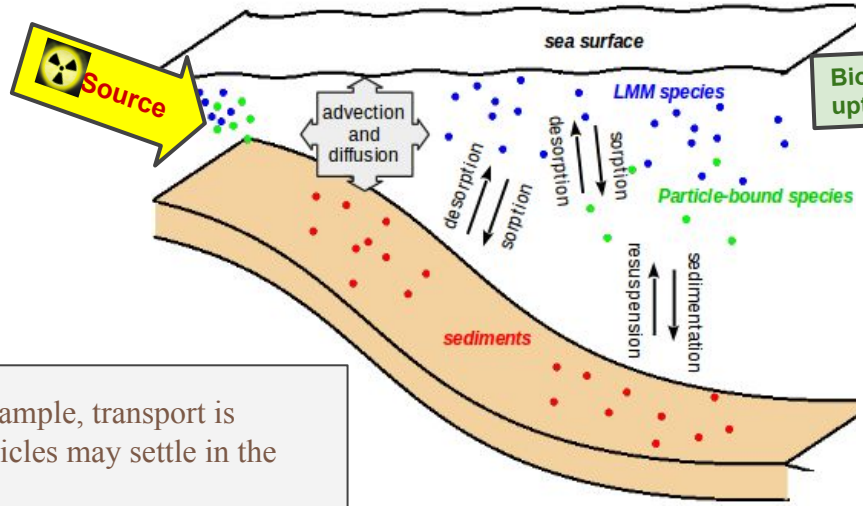
$$\frac{\partial A_w}{\partial t} = -k_1 A_w + k_2 A_s$$
$$\frac{\partial A_s}{\partial t} = k_1 A_w - k_2 A_s$$

Steady state (equilibrium) when t→∞

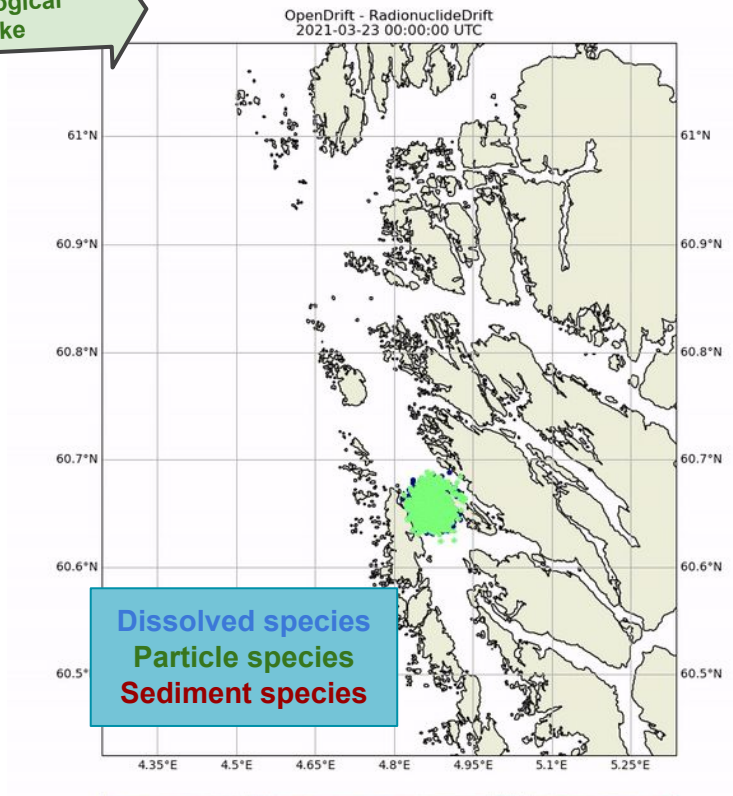




Dynamic transformations



Biological uptake



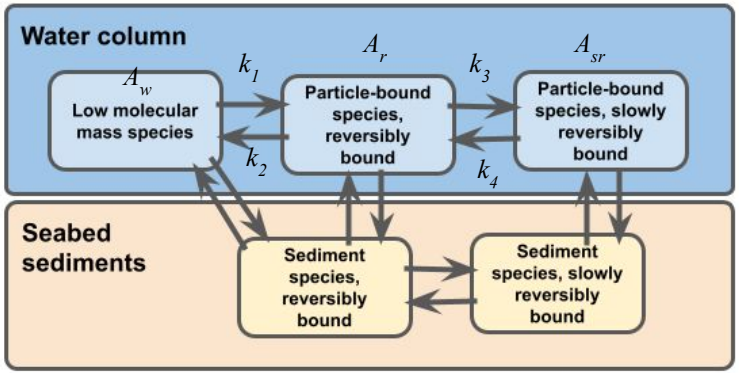
In a real-world example, transport is essential, and particles may settle in the sediments

Equilibrium, steady state, will not be the case

Dynamic transformations

- more complex speciation include slowly reversible bindings to particles

$$\frac{\partial A_w}{\partial t} = -k_1 A_w + k_2 A_r$$
$$\frac{\partial A_r}{\partial t} = k_1 A_w - k_2 A_r - k_3 A_r + k_4 A_{sr}$$
$$\frac{\partial A_{sr}}{\partial t} = k_3 A_r - k_4 A_{sr}$$



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Transport model example

https://github.com/magnesim/opendrift_simulations/blob/main/rad320_surface.py

Transport model
OpenDrift

Input
Release time
Discharge location
Currents
Settings

Output
Trajectories of particle drift

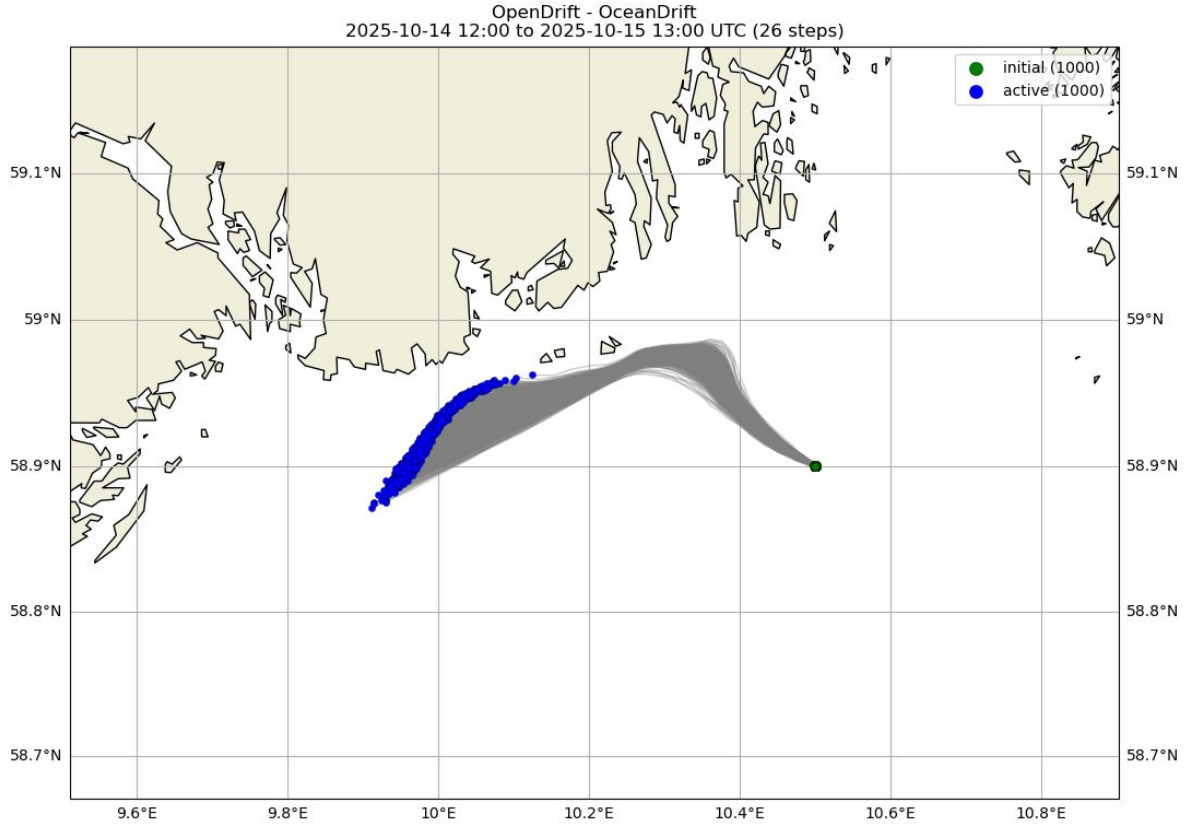
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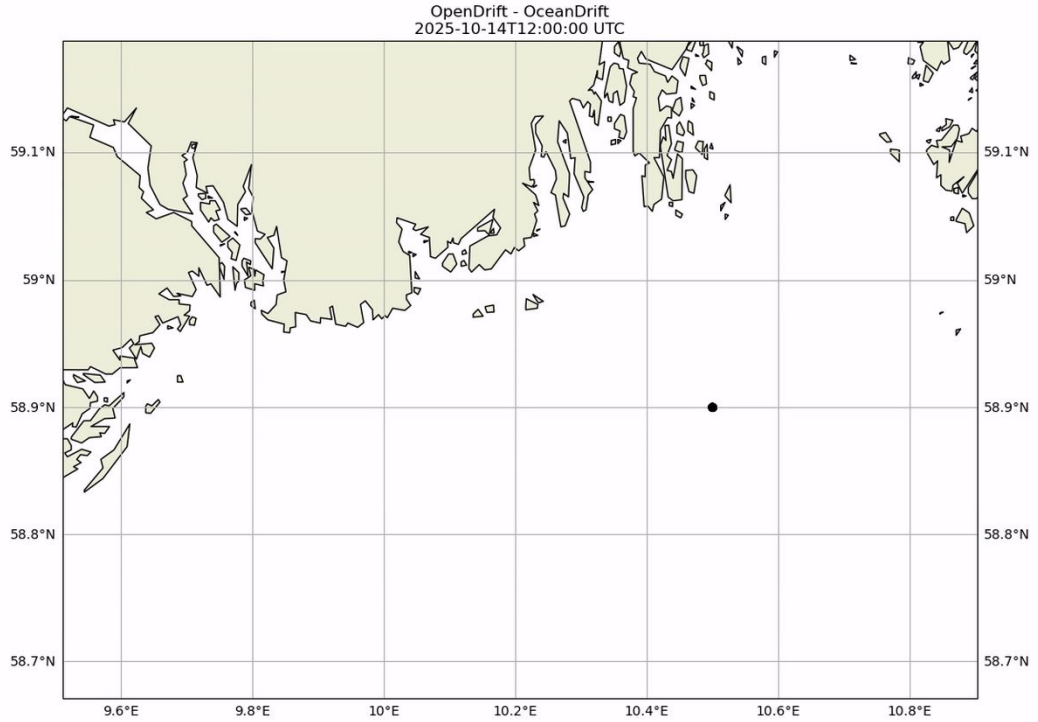
Transport model example

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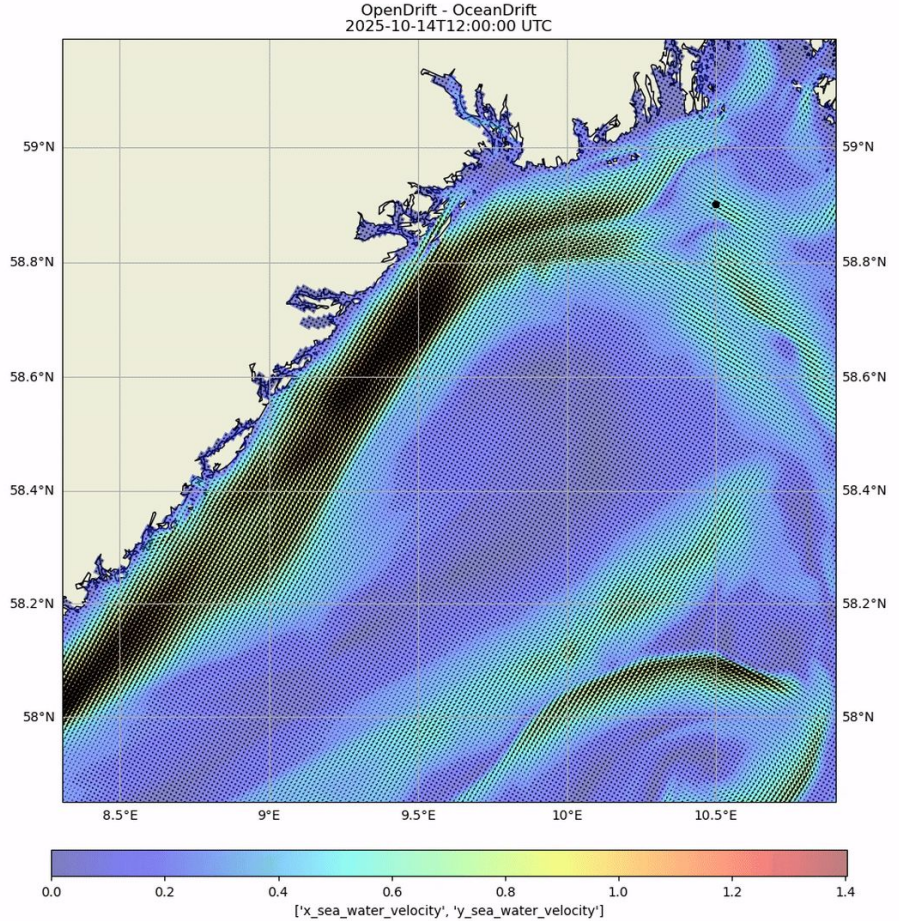
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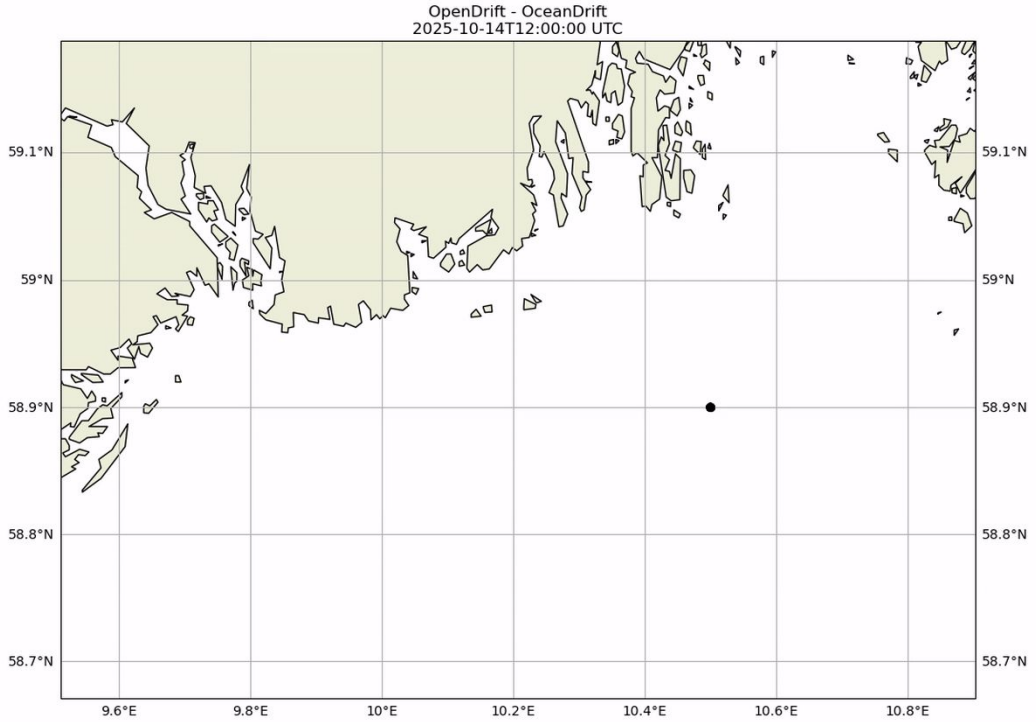


Compute concentrations

Concentrations can be computed by counting number of particles inside a box at a certain volume at a given time

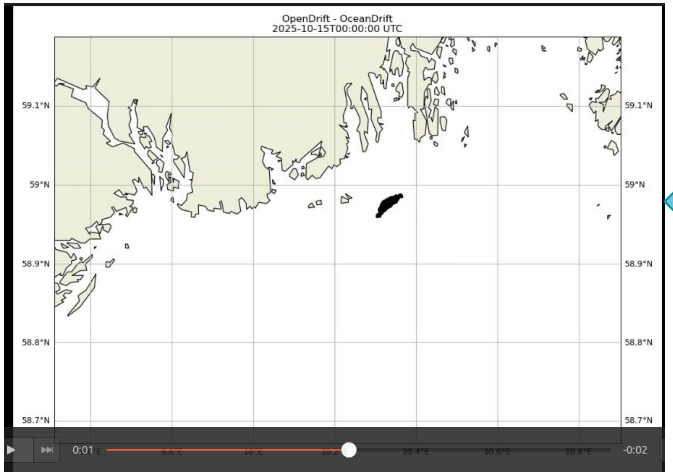
- create 3D grid
- for each time step, count number of particles inside each box
- multiply by weight (activity per particle)
- divide by volume

$$C = Nw / V$$

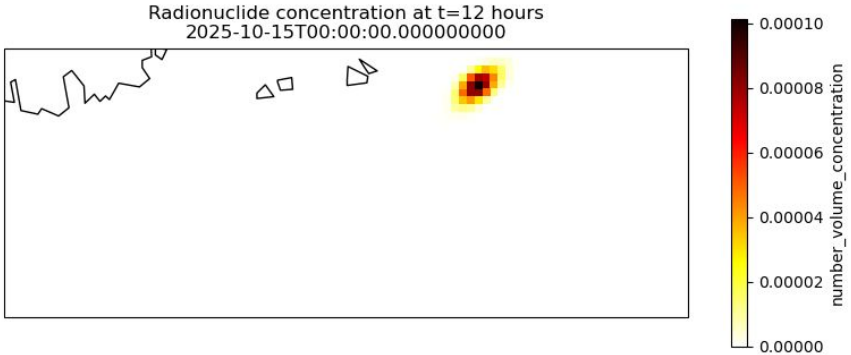


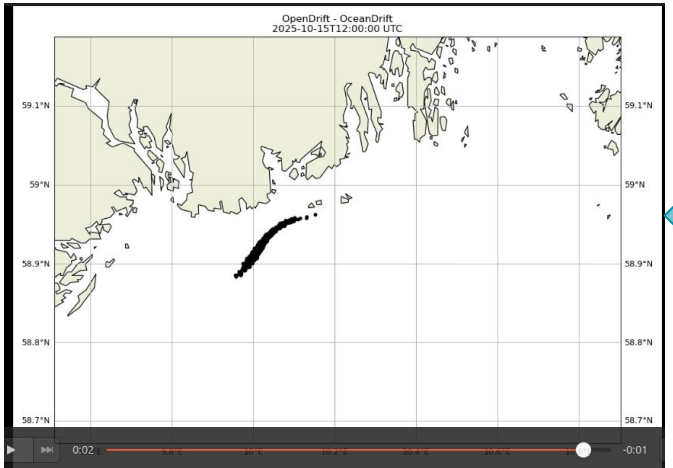
Particle locations

Concentrations in surface layer

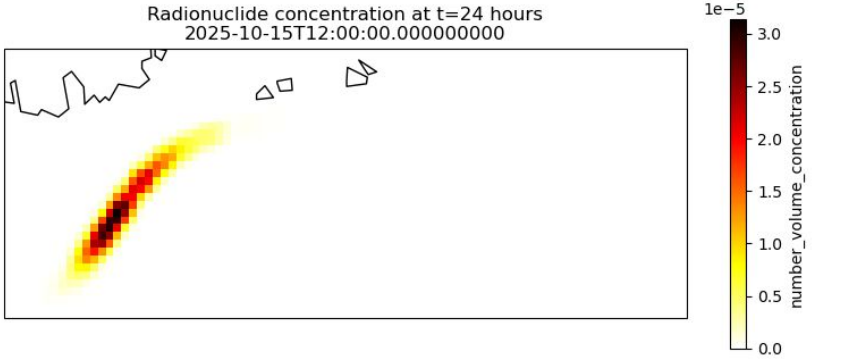


12 hours





24 hours



Summary

Marine environment
Radionuclides
Modeling

- The main driver for marine radionuclide transport is the ocean currents
- Ocean currents are highly variable and difficult to predict exactly
- Radionuclides can appear in a number of different forms/species
- The speciation (the distribution of species) affects the transport properties and the potential for biological uptake

- Models are simplified approximations to the real world
- The models need proper input (source, driver data, parameters)
- Physical processes (speciation, transformations) affects the transport estimates

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